

Lecture 9

The open Toda chain: integrability proved

Objectives. Everything in [Lectures 1 to 8](#), applied to one nonlinear many-body system. We prove that the open Toda chain of N particles has N integrals, independent at every point and in involution: the definition of [Lecture 4](#) is met in full. Then we check the hypotheses of the Liouville–Arnold theorem one by one, and one of them fails: the level sets are not compact, and they are not tori. What replaces the torus is scattering. The particles fly apart, their final momenta are the eigenvalues of the Lax matrix, and a three-body collision is a sequence of two-body ones — the classical fact that [Lecture 16](#) quantises.

9.1 FPUT, and the Toda potential

In 1955 Fermi, Pasta, Ulam and Tsingou put a chain of masses joined by weakly nonlinear springs on a computer, with spring energy

$$V_{\text{FPUT}}(r) = \frac{r^2}{2} + \alpha \frac{r^3}{3}, \quad (9.1)$$

r the extension of the spring. They started it with all the energy in the lowest normal mode and expected the nonlinearity to share it out among all the modes. It did not: the energy wandered into a few neighbouring modes and came back, almost completely ([Figure 1](#)). A system that behaves this regularly is behaving as if it had more integrals than energy.

In 1967 Toda looked for a spring law for which the chain has *exact* periodic nonlinear waves, trying elliptic functions as the nonlinear analogue of sines, and found the exponential (Toda, §2.1). With the constants set to one and $r_n = q_{n+1} - q_n$,

$$V_{\text{Toda}}(r) = e^{-r} + r - 1 = \frac{r^2}{2} - \frac{r^3}{6} + \frac{r^4}{24} - \cdots, \quad (9.2)$$

which to cubic order is the FPUT spring, with $\alpha = -\frac{1}{2}$ ([Figure 9.1](#)). Compressed ($r < 0$) it is a steep exponential wall; stretched, it softens to the constant force of the linear term.

For a *ring* of springs the linear terms add up to $\sum_n r_n =$ the circumference, a constant, and can be dropped; the result is the periodic chain of [Lecture 10](#), whose Poincaré section is [Figure 6.9](#). For an *open* chain nothing holds the ends, and we drop the linear terms by fiat.¹ What is left is the Hamiltonian of [Lecture 7](#),

$$H = \frac{1}{2} \sum_{n=1}^N p_n^2 + \sum_{n=1}^{N-1} e^{q_n - q_{n+1}}, \quad (9.3)$$

¹Toda, §3.10, following Moser: the end particles are sent to infinity, where the linear terms are an infinite constant.

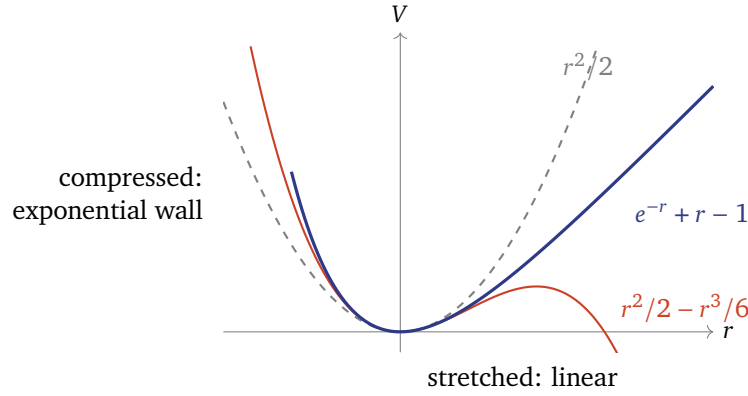


Figure 9.1: The Toda spring (9.2) (indigo), its harmonic approximation (dashed) and its cubic truncation, which is the FPUT spring with $\alpha = -\frac{1}{2}$ (red). All three agree to second order; the Toda spring is the one whose chain is exactly solvable.

N unit masses on a line, each repelled by its neighbours (Figure 9.2). The repulsion is felt only when particle n comes close to, or overtakes, particle $n + 1$. So the open chain has no FPUT oscillations: its particles fly apart. It is a nonlinear many-body system simple enough to take apart completely.

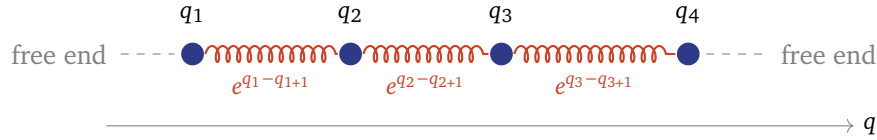


Figure 9.2: The open Toda chain, $N = 4$. Neighbours interact through $e^{q_n - q_{n+1}}$, which is small while particle n is well behind particle $n + 1$ and grows exponentially as it catches up. The end particles have one neighbour each.

9.2 Flaschka variables and the Lax pair, recalled

The phase space is $\mathbb{R}^{2N}$ with coordinates (q, p) , so $n = N$ degrees of freedom. § 7.6 put the equations of motion in Lax form, with Flaschka's variables (7.10), $a_n = e^{(q_n - q_{n+1})/2}$ and $b_n = p_n$, and

$$L = \begin{pmatrix} b_1 & a_1 & & \\ a_1 & b_2 & \ddots & \\ & \ddots & \ddots & a_{N-1} \\ & & a_{N-1} & b_N \end{pmatrix}, \quad M_{n,n+1} = -\frac{1}{2}a_n = -M_{n+1,n}, \quad \dot{L} = [M, L] \quad (9.4)$$

(Proposition 7.7). By Theorem 7.2 the eigenvalues of L are conserved, and so are the N functions

$$F_k = \text{tr } L^k, \quad k = 1, \dots, N, \quad (9.5)$$

which begin $F_1 = \sum_n p_n$, $F_2 = 2H$ (7.14). They are our candidates for Definition 4.6, with $H = \frac{1}{2}F_2$.

Three facts about L will be used over and over.

- (L1) *Simple spectrum.* Every $a_n > 0$, so L is a real symmetric (Jacobi) matrix with N distinct real eigenvalues $\lambda_1 < \dots < \lambda_N$ (proof of [Proposition 7.7](#)). Here λ_k are eigenvalues, not a spectral parameter: the open chain has none.
- (L2) *Bounded entries.* For a real symmetric matrix $|L_{ij}| \leq \max_k |\lambda_k|$, since $|L_{ij}| = |e_i \cdot L e_j|$ is at most the largest stretching factor of L . The eigenvalues are conserved, so along any motion every entry of L stays bounded by a constant fixed at $t = 0$.
- (L3) *Brackets.* Since $\{f, p_m\} = \partial f / \partial q_m$,

$$\{a_n, b_m\} = \frac{1}{2} a_n (\delta_{nm} - \delta_{m, n+1}), \quad \{a_n, a_m\} = \{b_n, b_m\} = 0. \quad (9.6)$$

The bond variable a_n is shifted only by the momenta at its two ends, with opposite signs.

9.3 Independence, by a Vandermonde argument

Three particles, by hand

For $N = 3$ the traces are $F_1 = \sum p_n$, $F_2 = \sum p_n^2 + 2(a_1^2 + a_2^2)$ and $F_3 = \sum p_n^3 + 3a_1^2(p_1 + p_2) + 3a_2^2(p_2 + p_3)$. Their derivatives with respect to the momenta alone form a 3×3 matrix with determinant

$$\det \frac{\partial(F_1, F_2, F_3)}{\partial(p_1, p_2, p_3)} = 6 \left[(p_2 - p_1)(p_3 - p_1)(p_3 - p_2) - a_1^2(p_2 - p_1) - a_2^2(p_3 - p_2) \right]. \quad (9.7)$$

When the particles are far apart, or their momentum *differences* large compared with the a_n (a common boost changes nothing), the a_n^2 terms are negligible, the F_k are nearly the power sums $\sum p_n^k$, and (9.7) is six times the *Vandermonde determinant* of the momenta: non-zero as soon as the momenta are distinct. That is why the far-apart (or large momentum-separation) regime is the right place to look first: there the chain is three free particles, and the three integrals are visibly three different functions.

But (9.7) is not a proof of anything everywhere. It vanishes when all three momenta are equal, and also at $p = (0, 1, 2)$, $a_1 = a_2 = 1$, where the momenta are distinct. At such points independence has to come from the q -derivatives. The general proof uses both, and is organised by L .

General N

Lemma 9.1 (Gradients of the traces). *With the conventions $a_0 = a_N = 0$,*

$$\frac{\partial F_k}{\partial p_n} = k (L^{k-1})_{nn}, \quad \frac{\partial F_k}{\partial q_n} = k \left[a_n (L^{k-1})_{n, n+1} - a_{n-1} (L^{k-1})_{n-1, n} \right]. \quad (9.8)$$

Proof. For any variable x , $\partial_x \operatorname{tr} L^k = k \operatorname{tr} (L^{k-1} \partial_x L)$, by the product rule and cyclicity (as in [Theorem 7.2\(i\)](#)). Now $\partial L / \partial p_n$ is the matrix unit E_{nn} , and a_n depends on q_n and q_{n+1} with $\partial a_n / \partial q_n = \frac{1}{2} a_n$, $\partial a_n / \partial q_{n+1} = -\frac{1}{2} a_n$. Each a_m sits in two symmetric slots and L^{k-1} is symmetric, which turns the two halves into one. ■

Lemma 9.2 (The band lemma). *Let L be a Jacobi matrix with all $a_n \neq 0$, and let X be a symmetric matrix commuting with L whose diagonal and first off-diagonals vanish. Then $X = 0$.*

Proof. Suppose not, and let $m \geq 2$ be the smallest distance from the diagonal at which X has a non-zero entry; write $x_i = X_{i,i+m}$. Compute the entry $(i, i+m-1)$ of $LX - XL$. Every entry of X closer than m to the diagonal is zero, and L is tridiagonal, so only one term survives from each product:

$$0 = [L, X]_{i,i+m-1} = a_{i-1} x_{i-1} - a_{i+m-1} x_i.$$

At $i = 1$ the first term is absent, and $a_m \neq 0$ gives $x_1 = 0$; then $i = 2$ gives $x_2 = 0$, and so on along the whole diagonal. So the m -th diagonal vanishes after all, a contradiction. ■

Both conditions on X are needed: $X = \mathbb{1}$ commutes with L and has a zero off-diagonal; and for $N = 2$ with $b_1 = b_2$, $X = E_{12} + E_{21}$ commutes with L and has a zero diagonal.

Theorem 9.3 (Independence). *The integrals $F_1, \dots, F_N$ of (9.5) are independent at every point of $\mathbb{R}^{2N}$.*

Proof. Suppose $\sum_k c_k dF_k = 0$ at some point. Put $g(\xi) = \sum_{k=1}^N k c_k \xi^{k-1}$, a polynomial of degree at most $N-1$, and $X = g(L)$, a symmetric matrix commuting with L . By (9.8), the p_n -component of $\sum_k c_k dF_k$ is X_{nn} and the q_n -component is $a_n X_{n,n+1} - a_{n-1} X_{n-1,n}$. All vanish. The first set says the diagonal of X is zero. The second, read at $n = 1, 2, \dots$ in turn, says $a_1 X_{12} = 0$, then $a_2 X_{23} = a_1 X_{12} = 0$, and so on: the first off-diagonal is zero. By Lemma 9.2, $X = 0$.

So $g(\lambda_j) = 0$ at each of the N distinct eigenvalues (L1). In the coefficients c_k this is N linear equations whose matrix $(k\lambda_j^{k-1})$ has determinant $N! \prod_{i < j} (\lambda_j - \lambda_i) \neq 0$, the Vandermonde determinant again. Hence every $c_k = 0$. ■

The Vandermonde determinant of the $N = 3$ example reappears at the end, with the momenta replaced by the eigenvalues, which are distinct at every point. That upgrades “independent where the momenta are distinct” to “independent everywhere”. Definition 4.1 asks only for a dense open set; § 9.5 needs the stronger statement.

9.4 Involution, from the Toda r -matrix

Theorem 8.7 gives $\{\text{tr } L^j, \text{tr } L^k\} = 0$ as soon as L satisfies a Sklyanin bracket (8.7). The open chain’s L has no spectral parameter, so we look for a *constant* matrix r on $\mathbb{R}^N \otimes \mathbb{R}^N$ with

$$\{L_1, L_2\} = [r, L_1 + L_2]. \quad (9.9)$$

The left-hand side

By (8.3) the entry of $\{L_1, L_2\}$ in row (a, c) and column (b, d) is $\{L_{ab}, L_{cd}\}$. By (9.6) it is non-zero only when one of the two entries is a diagonal b_m and the other is an a_n in slot $(n, n+1)$ or $(n+1, n)$.

Tutorial

For $N = 2$, with $a = a_1$, the sixteen entries collapse to

$$\{L_1, L_2\} = \frac{a}{2} \left[(E_{12} + E_{21}) \otimes (E_{11} - E_{22}) - (E_{11} - E_{22}) \otimes (E_{12} + E_{21}) \right] :$$

eight non-zero entries, each $\pm a/2$. Write out the 4×4 matrix in § 8.2’s ordering and compare it, entry by entry, with $[r, L_1 + L_2]$ for $r = \frac{1}{2}(E_{21} \otimes E_{12} - E_{12} \otimes E_{21})$ before reading the general proof.

The right-hand side, and the match

Proposition 9.4 (The open Toda r -matrix). *The constant matrix*

$$r = \frac{1}{2} \sum_{i < j} (E_{ji} \otimes E_{ij} - E_{ij} \otimes E_{ji}) \quad (9.10)$$

satisfies (9.9) for the Toda L . Consequently $\{F_j, F_k\} = 0$ for all j, k .

Proof. Put $\epsilon_{ij} = +1$ if $i < j$, -1 if $i > j$, 0 if $i = j$. Then $r = \frac{1}{2} \sum_{i,j} \epsilon_{ij} E_{ji} \otimes E_{ij}$, and

$$[r, L_1 + L_2] = \frac{1}{2} \sum_{i,j} \epsilon_{ij} \left([E_{ji}, L] \otimes E_{ij} + E_{ji} \otimes [E_{ij}, L] \right).$$

Expanding $[E_{ji}, L] = \sum_k (L_{ik} E_{jk} - L_{kj} E_{ki})$ and collecting the coefficient of $E_{ab} \otimes E_{cd}$ gives, for any matrix L ,

$$[r, L_1 + L_2]_{(ac),(bd)} = \frac{1}{2} (\epsilon_{ab} + \epsilon_{cd}) (L_{cb} \delta_{ad} - L_{ad} \delta_{bc}). \quad (9.11)$$

Compare with $\{L_{ab}, L_{cd}\}$, case by case.

- $c = d$, a diagonal entry. Then $\epsilon_{cd} = 0$ and (9.11) is $\frac{1}{2} \epsilon_{ab} L_{ab} (\delta_{ac} - \delta_{bc})$. For $(a, b) = (n, n+1)$ this is $\frac{1}{2} a_n (\delta_{nc} - \delta_{n+1,c}) = \{a_n, b_c\}$; for $(n+1, n)$ both signs flip and it is the same. For $|a - b| \geq 2$, $L_{ab} = 0$, as is the bracket; for $a = b$, both are zero.
- $a = b$. The same computation with the roles exchanged gives $-\{L_{cd}, b_a\} = \{b_a, L_{cd}\}$.
- *Both off-diagonal.* The bracket is zero. In (9.11) the prefactor vanishes unless both slots lie above the diagonal or both below. Say $a < b$ and $c < d$. If $a = d$ then $c < a < b$, so $|c - b| \geq 2$ and $L_{cb} = 0$; if $b = c$ then $a < b < d$ and $L_{ad} = 0$. Below the diagonal is the same.

So (9.9) holds entry by entry, and Theorem 8.7 gives the involution. ■

The model entered only in the last case: (9.11) holds for every matrix, and it matches the brackets because L is *tridiagonal*. (scripts/check-l9-algebra.py verifies (9.11) and (9.9) symbolically for $N \leq 5$.)

Not the CYBE: L8's “sufficient, not forced”, cashed

r is skew, $PrP = -r$, which is (8.8) for a constant r ; so Theorem 8.11 applies to it. Does it solve the classical Yang–Baxter equation (8.13)? **No.** Its left-hand side is

$$[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = -\frac{1}{4} [P_{12}, P_{13}] \neq 0. \quad (9.12)$$

This identity is *cited*, not proved here: Babelon–Bernard–Talon, eq. (4.32), state it in operator form, and scripts/check-l9-algebra.py checks it symbolically for $N = 2, 3$. An equation of this shape, with a fixed non-zero right-hand side in place of the zero, is the *modified* CYBE that § 8.6 named in passing.

And yet the Jacobi identity holds, as it must: the bracket is the canonical one on $\mathbb{R}^{2N}$. Theorem 8.11 asks only that the left of (9.12) commute with $L_1 + L_2 + L_3$, and it does, for *every* matrix L : P_{12} commutes with $L_1 + L_2$ (it only swaps them) and with L_3 (another leg), so it commutes with the sum, and so does P_{13} , and so does their commutator. This is Remark 8.13 in a real model: the CYBE is sufficient for Jacobi, and the Toda chain satisfies Jacobi without it.

Reading

How unique is r ? Adding $x\mathbb{1} + yP$ changes nothing in (9.9), since both commute with $L_1 + L_2$ (P only swaps the two legs). Babelon–Bernard–Talon, §4.7, derive (9.10) rather than check it, and show it is the only skew constant solution; `scripts/check-rmatrix.py` (V8) finds, at sampled points for $N = 3$, no constant solutions beyond $r + x\mathbb{1} + yP$. Of these, the skew one is r itself, which fails the CYBE, while $r \pm \frac{1}{2}P$ do solve the CYBE (Babelon–Bernard–Talon, (4.33); checked for $N = 2, 3$) and are not skew, so Theorem 8.11 as proved does not cover them. So no constant r for Toda is both skew and a solution of the CYBE: L8’s route “skew plus CYBE, hence Jacobi” is closed here, and Jacobi holds by the weaker condition above. That is not a defect of the model; it is why Lie-bialgebra theory has a modified equation.

9.5 The hypothesis audit: what we have, and what we have not

Theorem 9.5 (The open Toda chain is Liouville integrable). *On $\mathbb{R}^{2N}$, the functions $F_k = \text{tr } L^k$, $k = 1, \dots, N$, are independent at every point and in involution, and $H = \frac{1}{2}F_2$. The open Toda chain satisfies Definition 4.6.*

Proof. Theorem 9.3, Proposition 9.4, and (7.14). ■

That is the definition. Now the theorem. Theorem 4.8 is about a connected component Σ of a level set $\mathcal{M}_f = \{F = f\}$. Its hypotheses, and the places its proof spends them:

hypothesis	open Toda	where § 4.4 spends it
involution	✓ Proposition 9.4	stages (i)–(iii)
(I) independence on Σ	✓ Theorem 9.3, everywhere	stages (i), (iii), (iv)
Σ connected	✓ by choice of Σ	stage (iii)
(C) Σ compact	✗ fails for every Σ	stage (ii) completeness; stage (iv) full rank

Compactness fails. The flow of $F_1 = \sum_n p_n$ is $q_n \mapsto q_n + s$, $p_n \mapsto p_n$: a rigid translation of the whole chain. It changes no a_n and no b_n , so it preserves every F_k and maps Σ into itself, and its orbits are unbounded lines. A compact set contains no unbounded line. No level set of the open chain is compact, and Theorem 4.8 does not apply to any of them.

That does not end the audit. § 4.4 used compactness in two places, and one of them can be repaired by hand.

Proposition 9.6 (Completeness). *For any constants $t_1, \dots, t_N$, the Hamiltonian vector field of $G = \sum_k t_k F_k$ is complete. In particular the flows of $F_1, \dots, F_N$ are.*

Proof. The components of X_G are $\partial G / \partial p_n$ and $-\partial G / \partial q_n$, which by (9.8) are polynomials in the entries of L . Along the flow of G each F_k is conserved, since $\{F_k, G\} = \sum_j t_j \{F_k, F_j\} = 0$; the F_k fix the characteristic polynomial of L , so its spectrum is constant along an integral curve, and by (L2) the entries of L are bounded on it by a constant. Hence $|X_G| \leq C$ along the curve, which in a time T moves at most a distance CT and so stays in a compact ball. By the escape argument in the proof of Proposition 1.8, a curve that exists only up to a finite time must leave every compact set. So the curve exists for all time. ■

With completeness in hand, stages (i)–(iii) of § 4.4 and the first half of stage (iv) go through word for word: they use involution, (I), connectedness, and complete flows, and nothing else.

Corollary 9.7 (What survives · proved modulo Lemma 4.11 and Theorem 1.10). Every connected component Σ of every level set of $(F_1, \dots, F_N)$ is a Lagrangian submanifold of $\mathbb{R}^{2N}$ diffeomorphic to $T^k \times \mathbb{R}^{N-k}$ for some $k < N$, and the flows of the F_k are straight lines in the corresponding coordinates.

Proof. Stage (iv) with a discrete stabiliser gives $T^k \times \mathbb{R}^{N-k}$ for some $0 \leq k \leq N$ (Lemma 4.11); $k = N$ would make Σ compact. ■

The torus conclusion of Theorem 4.8 is not claimed for the open chain, and it is false. § 9.8 shows that $k = 0$: every fibre is a copy of $\mathbb{R}^N$, with no periodic direction at all. If this feels like a let-down after five lectures of tori, it is the point. The definition of integrability is algebraic (Definition 4.6); the torus is a *theorem*, with a hypothesis, and here the hypothesis fails. What the open chain does on its non-compact fibres is the subject of the rest of this lecture: it scatters.

9.6 Two particles in closed form

Take $N = 2$: $H = \frac{1}{2}(p_1^2 + p_2^2) + e^{q_1 - q_2}$. The mean momentum $\bar{p} = \frac{1}{2}(p_1 + p_2)$ is conserved, and the centre of mass $\frac{1}{2}(q_1 + q_2)$ moves at the constant speed $\bar{p}$. (We write $\bar{p}$ rather than the usual P , which in this lecture is the permutation operator.) The relative coordinate $x = q_1 - q_2$ obeys

$$\dot{x} = p_1 - p_2, \quad \ddot{x} = \dot{p}_1 - \dot{p}_2 = -2e^x,$$

and since $\frac{1}{2}(p_1^2 + p_2^2) = \bar{p}^2 + \frac{1}{4}(p_1 - p_2)^2$, the energy of the relative motion is

$$H - \bar{p}^2 = \frac{\dot{x}^2}{4} + e^x =: \frac{v^2}{4}, \quad v > 0. \quad (9.13)$$

It is positive because e^x is, and it fixes v . Particle 1 starts behind particle 2 ($x \rightarrow -\infty$, where the potential is negligible) and approaches at relative speed v ; the wall e^x stops it at $e^x = v^2/4$ and sends it back.

Proposition 9.8 (The two-particle chain). Every solution with $N = 2$ is, for some t_0 and a choice of origin,

$$x(t) = \ln \frac{v^2}{4} - 2 \ln \cosh \frac{v(t - t_0)}{2}, \quad q_{1,2}(t) = \bar{p}t \pm \frac{x(t)}{2}. \quad (9.14)$$

Its properties:

- (a) $\dot{x} = -v \tanh \frac{v(t - t_0)}{2}$ runs from $+v$ to $-v$: the momenta $p_{1,2} = \bar{p} \pm \frac{1}{2}\dot{x}$ are exchanged, from $\bar{p} + \frac{v}{2}$, $\bar{p} - \frac{v}{2}$ to $\bar{p} - \frac{v}{2}$, $\bar{p} + \frac{v}{2}$.
- (b) At every time the eigenvalues of L are $\lambda_{1,2} = \bar{p} \mp \frac{v}{2}$: the asymptotic momenta, and $v = \lambda_2 - \lambda_1$.
- (c) With $t_0 = 0$, as $t \rightarrow -\infty$ and $t \rightarrow +\infty$,

$$q_1 \simeq \lambda_2 t + \ln v \text{ (in)}, \quad q_2 \simeq \lambda_2 t - \ln v \text{ (out)},$$

and the same with λ_1 and the signs of $\ln v$ reversed for the other pair. The straight line carrying the momentum λ_2 is displaced by $-2 \ln v$, the one carrying λ_1 by $+2 \ln v$.

Proof. Differentiating (9.14), $\ddot{x} = -\frac{v^2}{2} \operatorname{sech}^2 \frac{v(t-t_0)}{2} = -2e^x$, and $\frac{1}{4}\dot{x}^2 + e^x = \frac{v^2}{4}(\tanh^2 + \operatorname{sech}^2) = \frac{v^2}{4}$. Every solution has exactly one turning point: $\ddot{x} < 0$, and $\dot{x}$ cannot stay positive, since then x increases and $\ddot{x} \leq -2e^{x(0)}$; likewise backwards in time. A solution is fixed by its energy and the time t_0 of its turning point, where $x = \ln(v^2/4)$ and $\dot{x} = 0$, and (9.14) has both free. (a) follows. For (b), $\operatorname{tr} L = 2\bar{p}$ and $\det L = p_1 p_2 - e^x = \bar{p}^2 - \frac{1}{4}\dot{x}^2 - e^x = \bar{p}^2 - \frac{1}{4}v^2$ by (9.13), so the characteristic polynomial is $(\eta - \lambda_1)(\eta - \lambda_2)$ for all t . For (c), use $\ln \cosh u = |u| - \ln 2 + o(1)$: then $x = \ln v^2 - v|t| + o(1)$, and $q_1 = \bar{p}t + \frac{x}{2} \simeq (\bar{p} + \frac{v}{2})t + \ln v$ as $t \rightarrow -\infty$, $q_2 = \bar{p}t - \frac{x}{2} \simeq (\bar{p} - \frac{v}{2})t - \ln v$ as $t \rightarrow +\infty$. ■

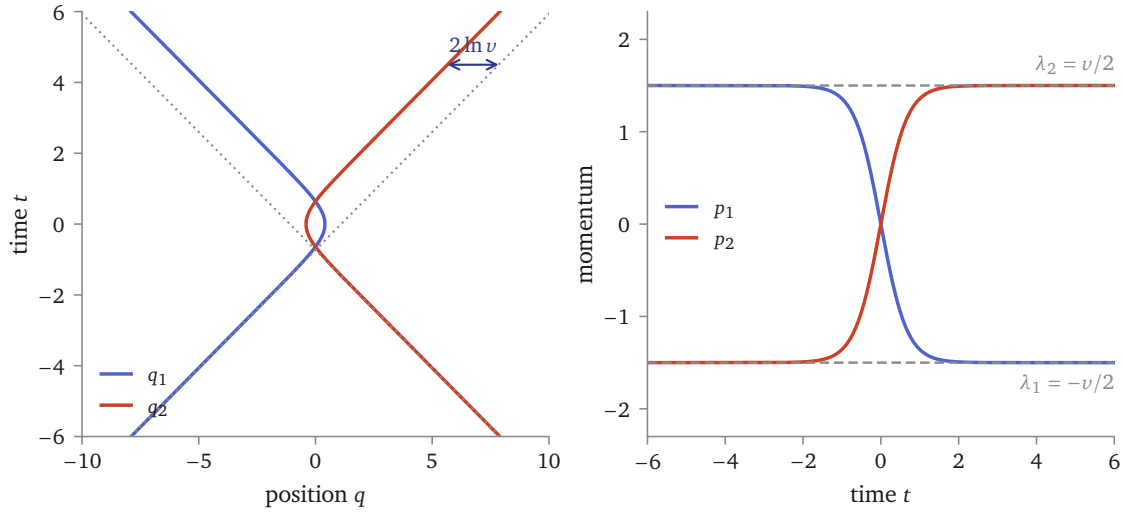


Figure 9.3: Two Toda particles, centre-of-mass frame, $v = 3$. Left: world lines (time upward). The dotted lines continue the *incoming* straight lines; after the collision each momentum travels on the other particle, on a line parallel to its incoming one and displaced by $2 \ln v$. Right: the momenta are exchanged, and the eigenvalues of L (dashed) are the asymptotic momenta $\pm v/2$ at every time (scripts/figures/toda_two_body.py, which checks (9.14) against a direct integration and the spectrum at every step).

Read Figure 9.3 as a scattering experiment. In the far past there are two free particles with momenta $\lambda_2 > \lambda_1$, the faster one behind; in the far future, two free particles with the *same two momenta*, the faster one in front. Following the momentum rather than the particle, it is as if they had passed through each other, except that each straight line is displaced: the fast one by $-2 \ln v$, a time delay $2 \ln v / \lambda_2$ when $\lambda_2 > 0$. This displacement is the classical shadow of the two-body phase shift of Lecture 16. For $v < 1$ its sign reverses: slow particles turn round while still far apart, and the momentum is handed forward across the gap.

9.7 General N : elastic, non-diffractive scattering

For N particles the general solution has determinant formulas (Moser; Babelon–Bernard–Talon, §4.8), which we omit: the asymptotics are all we need.

Theorem 9.9 (Moser · cited). *For every solution of the open Toda chain, $q_{n+1}(t) - q_n(t) \rightarrow +\infty$ as $t \rightarrow \pm\infty$, and each particle moves asymptotically on a straight line: $p_n(t) \rightarrow p_n^\pm$ and*

$q_n(t) = p_n^\pm t + \beta_n^\pm + o(1)$. With $\lambda_1 < \dots < \lambda_N$ the eigenvalues of L ,

$$p_n^+ = \lambda_n, \quad p_n^- = \lambda_{N+1-n}. \quad (9.15)$$

The line carrying λ_k arrives on particle $N+1-k$ and leaves on particle k , displaced by

$$D_k := \beta_k^+ - \beta_{N+1-k}^- = \sum_{j>k} 2 \ln(\lambda_j - \lambda_k) - \sum_{j<k} 2 \ln(\lambda_k - \lambda_j). \quad (9.16)$$

Given the separation, (9.15) is one line. If $q_{n+1} - q_n \rightarrow \infty$ then every $a_n \rightarrow 0$ and $L(t) \rightarrow \text{diag}(p^\pm)$. Eigenvalues depend continuously on the matrix and do not change in time, so $\{p_n^\pm\}$ is the spectrum $\{\lambda_k\}$. The separation fixes the order: $q_{n+1} - q_n \rightarrow +\infty$ as $t \rightarrow +\infty$ needs $p_{n+1}^+ \geq p_n^+$, so the outgoing momenta increase along the chain, and as $t \rightarrow -\infty$ they decrease. This is what the eigenvalues of L mean, which is what Figure 7.6 was hinting at: they are the momenta the particles have when they are free. The separation and (9.16) are Moser's (1975, pp. 480–481, in his normalisation; Toda, §3.10, gives (9.15) with L halved); `scripts/check-19-algebra.py` integrates $N = 3, 4$ in ours and finds both to within 10^{-8} .

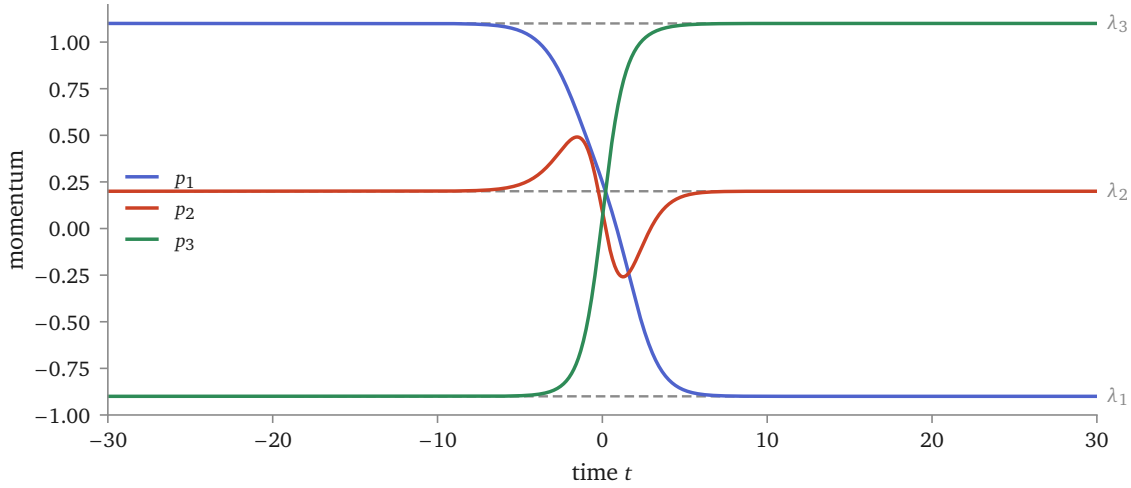


Figure 9.4: Three particles scatter. Their momenta (solid) start on the eigenvalues of L (dashed) in decreasing order and end on them in increasing order; in between, the eigenvalues do not move (`scripts/figures/toda_scattering.py`, which asserts the spectrum constant to 10^{-9} and both limits).

Two words describe this (Figure 9.4). The scattering is *elastic*: the set of outgoing momenta equals the set of incoming ones, so each particle's kinetic energy is handed on intact rather than shared. And it is *non-diffractive*: the outgoing momenta are not merely constrained by energy and momentum, which for $N \geq 3$ would allow a continuum of outcomes; they are exactly the incoming ones, permuted. N conserved eigenvalues leave room for nothing else.

Three particles: collisions in either order

Look at (9.16) term by term. The line carrying λ_k picks up $+2 \ln(\lambda_j - \lambda_k)$ from every faster momentum and $-2 \ln(\lambda_k - \lambda_j)$ from every slower one: exactly the two-body displacements of Proposition 9.8, one per pair, added. The total depends on the eigenvalues only. In particular it does not depend on the *order* in which the pairs met.

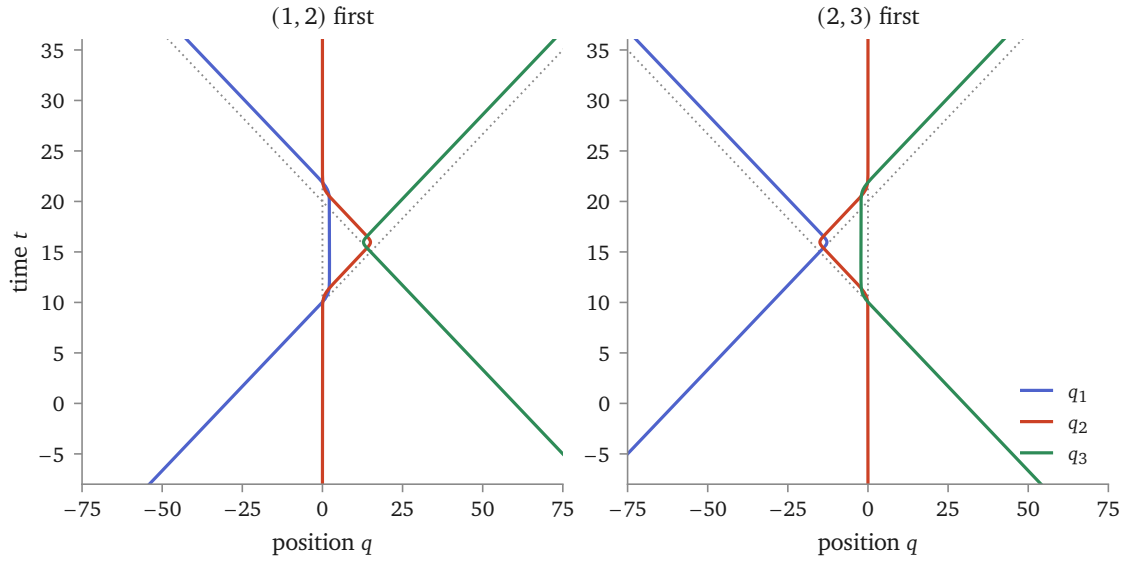


Figure 9.5: Three particles with the same incoming momenta 3, 0, -3 and different starting positions. Left, particles 1 and 2 meet first; right, 2 and 3 do. Each run is three two-body collisions, in different orders. The dotted lines continue the incoming lines; the net displacements are the same in both runs, $D = (2 \ln 18, 0, -2 \ln 18)$, as (9.16) says (`scripts/figures/toda_three_body.py`, which asserts it; the check script also runs a genuine three-body collision and gets the same).

Figure 9.5 makes it concrete. With incoming momenta 3, 0, -3 on particles 1, 2, 3, the collisions exchange momenta pairwise until the order is reversed. It takes three exchanges, and they can happen in the order (12), (23), (12) or (23), (12), (23). The two runs start from different incoming lines and look different throughout, but each outgoing line is displaced from its own incoming line by the same amount in both. Written as permutations of the three momentum lines, $s_1 s_2 s_1 = s_2 s_1 s_2$ (Figure 9.6), and the displacements respect it.

This is the concrete content of the phrase *factorised scattering*: the N -body scattering map is a product of two-body maps, and the product does not depend on how it is bracketed in time. The general theorem, that every N -body Toda collision decomposes this way, is Moser's · *cited*; what we have seen is the two-body map in closed form and the three-body composition checked. Lecture 16 quantises exactly this: the two-body displacement becomes a two-body S-matrix, and the braid relation becomes the Yang–Baxter equation.

9.8 Why the fibre is non-compact

Corollary 9.7 left the number k of periodic directions open. The scattering picture settles it.

Proposition 9.10 (The fibres are planes · *sketched*). In Corollary 9.7, $k = 0$: every Σ is diffeomorphic to $\mathbb{R}^N$.

Proof (sketch). A stabiliser element $t \neq 0$ means that the flow Φ of $G = \sum_k t_k F_k$ returns every point of Σ to itself at time 1; in particular it returns the points $x(t)$ of a trajectory at late times. There, by Theorem 9.9, the particles are far apart and $L \simeq \text{diag}(\lambda)$. By (9.8) X_G then moves particle n at the speed $\partial G / \partial p_n \simeq g(\lambda_n)$, $g(\xi) = \sum_k k t_k \xi^{k-1}$; its speed is bounded on Σ (proof of Proposition 9.6), so in unit time the particles stay far apart and the momenta

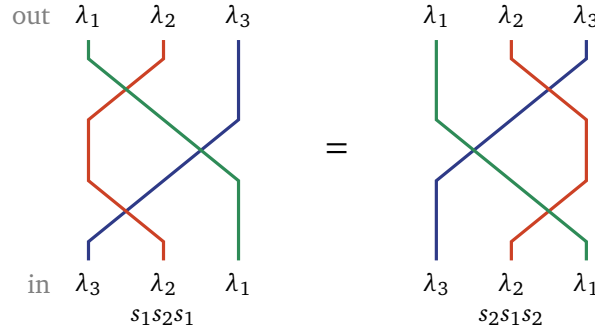


Figure 9.6: The three-body scattering of Figure 9.5 as a permutation of momentum lines, time upward. Each crossing is a two-body collision, which exchanges two adjacent momenta and displaces both lines by the two-body amount of Proposition 9.8. The two orders are the braid relation $s_1s_2s_1 = s_2s_1s_2$, and they carry the same total displacement on each line. The same picture with lines for particles and crossings for R -matrices is the Yang–Baxter equation of Lecture 17.

barely change. Returning in time 1 forces $g(\lambda_n) = 0$ for every n , hence $g = 0$ and $t = 0$ by the Vandermonde argument of Theorem 9.3. The stabiliser is trivial. ■

On a torus the commuting flows have a full lattice of periods, even though a single irrational flow need never return (Figure 4.2). On the open chain's fibre no combination of the flows ever returns; the proof used the physical flow of H , whose late-time configurations are free particles moving apart. In effect $\Sigma \cong \mathbb{R}^N$ is coordinatised by the incoming offsets β_n^- of Theorem 9.9, while the asymptotic momenta are fixed on it (Figure 9.7). They are the integrals, and they are also the obstruction: as long as they are distinct, the particles separate.

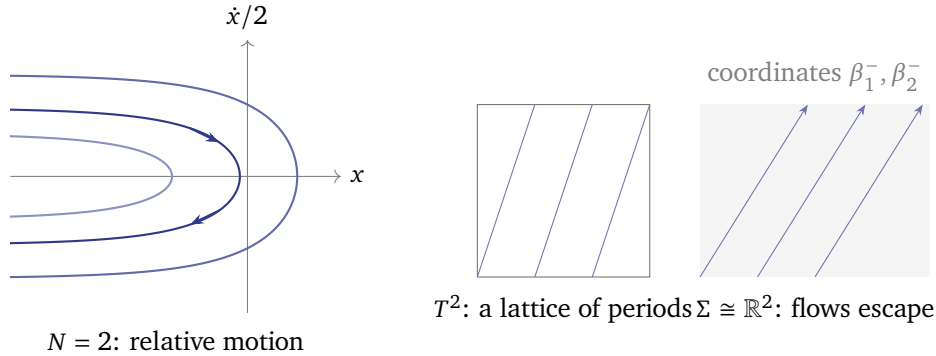


Figure 9.7: Non-compact fibres. Left: for $N = 2$ the relative motion lies on the curves $\frac{1}{4}\dot{x}^2 + e^x = v^2/4$ of (9.13), three of which are drawn. Each is a line open to $x \rightarrow -\infty$, run through once: the particles approach, turn and separate. Together with the free centre-of-mass line this makes each fibre a plane. Right: the contrast with Theorem 4.8. On a torus the commuting flows wrap round, with a full lattice of periods; on the open chain's fibre they are straight lines in the incoming offsets and never return (Proposition 9.10).

Close the chain into a ring and this changes. The whole ring can still drift (the flow of F_1 is still a translation), but the particles can no longer separate. Once the total momentum is fixed and the common translation split off, the relative level sets are compact, and each

regular connected component is a torus T^{N-1} , whose actions are read off a curve. That is [Lecture 10](#).

Summary.

- *The model*: the Toda spring $e^{-r} + r - 1$ is FPUT's to cubic order; the open chain (9.3) is pure nearest-neighbour repulsion, with integrals $F_k = \text{tr } L^k$, $k = 1, \dots, N$.
- *Independence everywhere* ([Theorem 9.3](#)): gradients of traces, the band lemma, and the Vandermonde determinant of the distinct eigenvalues.
- *Involution* ([Proposition 9.4](#)): a constant skew r with $\{L_1, L_2\} = [r, L_1 + L_2]$; it solves the modified CYBE, not the CYBE, and Jacobi holds anyway.
- *The audit* ([Theorem 9.5](#) and [Corollary 9.7](#)): Liouville integrable, flows complete, no fibre compact. Each fibre is $\mathbb{R}^N$ ([Proposition 9.10](#), sketched); the torus conclusion is false here.
- *Scattering*: two particles exchange momenta, each line displaced by $\mp 2 \ln(\lambda_2 - \lambda_1)$ ([Proposition 9.8](#)). For N (Moser, cited) the asymptotic momenta are the eigenvalues of L , reversed; the scattering is elastic and non-diffractive, and the displacements add up pairwise, whatever the collision order.

Tutorial. Exercise Sheet 9.

References

Toda, *Theory of Nonlinear Lattices*, §2.1 (finding the potential) and §3.10 (Moser's finite lattice); Babelon, Bernard and Talon, *Introduction to Classical Integrable Systems*, §4.6–4.8 (the Lax pair, the r -matrix and the modified Yang–Baxter equation, the solution); J. Moser, “Finitely many mass points on the line under the influence of an exponential potential — an integrable system”, *Lecture Notes in Physics* 38 (1975) 467, and *Adv. Math.* 16 (1975) 197; M. Hénon, *Phys. Rev. B* 9 (1974) 1921; H. Flaschka, *Phys. Rev. B* 9 (1974) 1924. **Conventions.** Toda's book halves L , so his asymptotic momenta are $2\lambda_k$; Babelon–Bernard–Talon's bracket $\{p, q\} = \frac{1}{2}$ reverses the sign of their r relative to (9.10).